never varies by conviction, volatility or stop distance.

07 One position at a time. No pyramiding, no averaging down, no re-entry while a position is open.

COSTS AND FILLS

- **No fees, no funding, no slippage.** No fees, funding or slippage. Stops fill exactly at the stop price.
- Stops are assumed to fill exactly at the stop price. A real gap through the level fills worse, so tight stops in particular would look **worse** in live trading, not better.
- Entries fill at an open; trend exits fill at a close. That asymmetry is inherited from the original run and is documented rather than quietly corrected — see Method & limits.

THE DATA

- **BTCUSDT, Binance, daily candles**
- **3,172 daily bars**, 2018-01-01 → 2026-09-07
- The 20/50 crossover produces **36 trades** over that window — roughly four a year.
- Sweep grid: **500 levels**, evenly spaced from 0.5% to 25%.

WHY THE ENTRIES ARE FROZEN

If entries and stops both moved, a better result could come from either, and neither could be attributed. Freezing the entry list turns the comparison into a controlled test with one variable — the same reason every version reports the same 36 entry dates.

STOP FIXED AT 2.022% BELOW ENTRY

TIGHT · 2% stop

RETURN	WIN RATE	MAX DRAWDOWN	STOPS FIRED	CAME BACK IN 3D
+325%	13.9%	-20.1%	31/36	71.0%
4.2x equity	5 of 36 trades	peak to trough, closed equity	86% of trades	22 of 31 stop-outs

ENTRY & EXIT

The rules for this version

ENTRY

01 Market: **BTCUSDT on Binance**, daily candles, UTC close. No other pair is traded.

02 Indicators: simple moving averages of the daily close, `SMA(20)` and `SMA(50)`.

03 Trigger: on the close of bar `t`, `SMA20` crosses **above** `SMA50` – it must be below on `t-1` and above on `t`.

04 Fill: buy at the **open of bar t+1**. The signal bar is never traded on its own close.

05 Direction: **long only.** No shorts, no leverage, no hedges.

06 Size: **100% of current equity**, compounded trade to trade. Size never varies by conviction, volatility or stop distance.

07 One position at a time. No pyramiding, no averaging down, no re-entry while a position is open.

EXIT – THE STOP

01 Stop price is fixed at entry: `stop = entry × (1 - 0.020220)` – a **2%** stop (exactly `2.022%`).

02 The stop is **static**. It never trails, never widens, never moves to breakeven.

03 Checked against each subsequent bar's **low**, starting at bar `t+2` – the entry bar itself is never stop-checked.

04 If `low ≤ stop`, the trade closes **at the stop price** on that bar. Gaps through the level are not modelled.

05 Whichever comes first – stop or trend exit – ends the trade.

EXIT – THE TREND

01 On the close of bar `k`, `SMA20` crosses **below** `SMA50`.

02 Fill at the **close of bar k+1**.

03 A position still open on the last bar of the dataset is closed at that bar's close.

THE CURVE

Equity, trade by trade



Marked at every closed trade. Final equity **4.2x** from 1.00x.

HOW FAR UNDERWATER, AND WHEN

Drawdown



Worst closed-equity drawdown **-20.1%**. This is measured on realised equity between trades, so it understates the intra-trade pain – for that, read the **Worst** column in the trade table.

ALL 36

Every trade



Each bar is one trade, scaled by its return. Above the line is a win; below is a loss, red where the stop ended it.

All 36 trades. **Worst** and **Best** are the deepest and highest the trade went while open, against the entry price — the drawdown you actually sat through.

1	2018-03-05	2018-03-06	1d	$11,515	$11,282	$11,282	STOP	-2.02%	0.98x	-8.3%	+1.7%
2	2018-04-27	2018-04-28	1d	$9,267	$9,080	$9,080	STOP	-2.02%	0.96x	-4.3%	+1.7%
3	2018-07-23	2018-08-04	12d	$7,395	$7,245	$7,245	STOP	-2.02%	0.94x	-6.3%	+14.8%
4	2018-10-23	2018-10-29	6d	$6,581	$6,448	$6,448	STOP	-2.02%	0.92x	-4.0%	+0.7%
5	2019-01-09	2019-01-10	1d	$3,955	$3,875	$3,875	STOP	-2.02%	0.90x	-10.5%	+1.3%
6	2019-02-22	2019-02-24	2d	$3,937	$3,858	$3,858	STOP	-2.02%	0.88x	-5.7%	+6.6%
7	2019-08-18	2019-08-21	3d	$10,216	$10,009	$10,009	STOP	-2.02%	0.87x	-3.5%	+7.2%
8	2019-11-06	2019-11-07	1d	$9,308	$9,120	$9,120	STOP	-2.02%	0.85x	-2.2%	+1.4%
9	2020-01-09	2020-01-10	1d	$8,055	$7,892	$7,892	STOP	-2.02%	0.83x	-4.8%	+1.8%
10	2020-04-20	2020-04-21	1d	$7,121	$6,977	$6,977	STOP	-2.02%	0.82x	-5.2%	+1.4%
11	2020-07-28	2020-08-02	5d	$11,030	$10,807	$10,807	STOP	-2.02%	0.80x	-4.6%	+9.9%

THE RECOVERY TEST

Did price come back?

For every trade the stop ended, price is checked for the next **3 trading days**: did the daily close get back above the stop level? A "yes" means the stop did not save the trade — it ended a position that was about to recover.

RESULT FOR THIS VERSION

22 of 31 stop-outs (71.0%) came back within 3 days.

STOP FIXED AT 9.976% BELOW ENTRY

WIDE • 10% stop

RETURN	WIN RATE	MAX DRAWDOWN	STOPS FIRED	CAME BACK IN 3D
+1,563%	44.4%	-36.8%	14/36	50.0%
16.6x equity	16 of 36 trades	peak to trough, closed equity	39% of trades	7 of 14 stop-outs

ENTRY & EXIT

The rules for this version

ENTRY

01 Market: **BTCUSDT on Binance**, daily candles, UTC close. No other pair is traded.

02 Indicators: simple moving averages of the daily close, `SMA(20)` and `SMA(50)`.

03 Trigger: on the close of bar `t`, `SMA20` crosses **above** `SMA50` – it must be below on `t-1` and above on `t`.

04 Fill: buy at the **open of bar t+1**. The signal bar is never traded on its own close.

05 Direction: **long only.** No shorts, no leverage, no hedges.

06 Size: **100% of current equity**, compounded trade to trade. Size never varies by conviction, volatility or stop distance.

07 One position at a time. No pyramiding, no averaging down, no re-entry while a position is open.

EXIT – THE STOP

01 Stop price is fixed at entry: `stop = entry × (1 - 0.099760)` – a **10%** stop (exactly `9.976%`).

02 The stop is **static**. It never trails, never widens, never moves to breakeven.

03 Checked against each subsequent bar's **low**, starting at bar `t+2` – the entry bar itself is never stop-checked.

04 If `low ≤ stop`, the trade closes **at the stop price** on that bar. Gaps through the level are not modelled.

05 Whichever comes first – stop or trend exit – ends the trade.

EXIT – THE TREND

01 On the close of bar `k`, `SMA20` crosses **below** `SMA50`.

02 Fill at the **close of bar k+1**.

03 A position still open on the last bar of the dataset is closed at that bar's close.

THE CURVE

Equity, trade by trade



Marked at every closed trade. Final equity **16.6x** from 1.00x.

HOW FAR UNDERWATER, AND WHEN

Drawdown



Worst closed-equity drawdown **-36.8%**. This is measured on realised equity between trades, so it understates the intra-trade pain – for that, read the **Worst** column in the trade table.

ALL 36

Every trade



Each bar is one trade, scaled by its return. Above the line is a win; below is a loss, red where the stop ended it.

All 36 trades. **Worst** and **Best** are the deepest and highest the trade went while open, against the entry price — the drawdown you actually sat through.

1	2018-03-05	2018-03-07	2d	$11,515	$10,366	$10,366	STOP	-9.98%	0.90x	-18.5%	+1.7%
2	2018-04-27	2018-05-11	14d	$9,267	$8,343	$8,343	STOP	-9.98%	0.81x	-10.0%	+8.1%
3	2018-07-23	2018-08-08	16d	$7,395	$6,657	$6,657	STOP	-9.98%	0.73x	-17.2%	+14.8%
4	2018-10-23	2018-10-30	7d	$6,581	$5,925	$6,331	TREND	-3.80%	0.70x	-4.0%	+0.7%
5	2019-01-09	2019-01-10	1d	$3,955	$3,561	$3,561	STOP	-9.98%	0.63x	-10.5%	+1.3%
6	2019-02-22	2019-07-30	158d	$3,937	$3,545	$9,574	TREND	+143.17%	1.54x	-6.8%	+254.8%
7	2019-08-18	2019-08-29	11d	$10,216	$9,197	$9,498	TREND	-7.02%	1.43x	-8.8%	+7.2%
8	2019-11-06	2019-11-17	11d	$9,308	$8,379	$8,379	STOP	-9.98%	1.29x	-10.3%	+1.4%
9	2020-01-09	2020-03-06	57d	$8,055	$7,251	$9,132	TREND	+13.37%	1.46x	-4.8%	+30.4%
10	2020-04-20	2020-06-30	71d	$7,121	$6,411	$9,139	TREND	+28.33%	1.87x	-5.2%	+45.8%
11	2020-07-28	2020-09-04	38d	$11,030	$9,930	$9,930	STOP	-9.98%	1.68x	-10.5%	+13.0%

THE RECOVERY TEST

Did price come back?

For every trade the stop ended, price is checked for the next **3 trading days**: did the daily close get back above the stop level? A "yes" means the stop did not save the trade — it ended a position that was about to recover.

RESULT FOR THIS VERSION

7 of 14 stop-outs (50.0%) came back within 3 days.

THE CONTROL — NO STOP AT ALL

NO STOP

RETURN	WIN RATE	MAX DRAWDOWN	STOPS FIRED
+648%	44.4%	-55.3%	0/36
7.5x equity	16 of 36 trades	peak to trough, closed equity	never

ENTRY & EXIT

The rules for this version

ENTRY

01 Market: **BTCUSDT on Binance**, daily candles, UTC close. No other pair is traded.

02 Indicators: simple moving averages of the daily close, `SMA(20)` and `SMA(50)`.

03 Trigger: on the close of bar `t`, `SMA20` crosses **above** `SMA50` – it must be below on `t-1` and above on `t`.

04 Fill: buy at the **open of bar t+1**. The signal bar is never traded on its own close.

05 Direction: **long only.** No shorts, no leverage, no hedges.

06 Size: **100% of current equity**, compounded trade to trade. Size never varies by conviction, volatility or stop distance.

07 One position at a time. No pyramiding, no averaging down, no re-entry while a position is open.

EXIT – THE STOP

01 **There is no stop.** The position is held through every drawdown until the trend exit fires.

02 This is the control. It isolates what the stop is doing by removing it entirely.

EXIT – THE TREND

01 On the close of bar `k`, `SMA20` crosses **below** `SMA50`.

02 Fill at the **close of bar k+1**.

03 A position still open on the last bar of the dataset is closed at that bar's close.

THE CURVE

Equity, trade by trade



Marked at every closed trade. Final equity **7.5x** from 1.00x.

HOW FAR UNDERWATER, AND WHEN

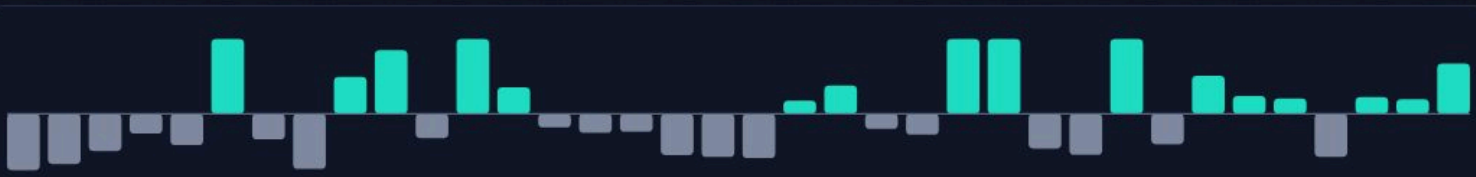
Drawdown



Worst closed-equity drawdown **-55.3%**. This is measured on realised equity between trades, so it understates the intra-trade pain – for that, read the **Worst** column in the trade table.

ALL 36

Every trade



Each bar is one trade, scaled by its return. Above the line is a win; below is a loss.

30	158d	$3,937	$9,574	TREND	+143.17%	1.09x	-6.8%	+254.8%
29	11d	$10,216	$9,498	TREND	-7.02%	1.01x	-8.8%	+7.2%
25	19d	$9,308	$7,110	TREND	-23.62%	0.77x	-30.0%	+1.4%
06	57d	$8,055	$9,132	TREND	+13.37%	0.88x	-4.8%	+30.4%
30	71d	$7,121	$9,139	TREND	+28.33%	1.12x	-5.2%	+45.8%
10	44d	$11,030	$10,337	TREND	-6.28%	1.05x	-10.9%	+13.0%

RULE TEST #02 · the full record BTCUSDT, Binance, daily candles

Personal experience. Not financial advice.

STOP FIXED AT 7.8156% BELOW ENTRY

BEST • ~7.8% stop

RETURN	WIN RATE	MAX DRAWDOWN	STOPS FIRED	CAME BACK IN 3D
+2,198%	44.4%	-30.5%	15/36	60.0%
23.0x equity	16 of 36 trades	peak to trough, closed equity	42% of trades	9 of 15 stop-outs

ENTRY & EXIT

The rules for this version

ENTRY

01 Market: **BTCUSDT on Binance**, daily candles, UTC close. No other pair is traded.

02 Indicators: simple moving averages of the daily close, `SMA(20)` and `SMA(50)`.

03 Trigger: on the close of bar `t`, `SMA20` crosses **above** `SMA50` – it must be below on `t-1` and above on `t`.

04 Fill: buy at the **open of bar t+1**. The signal bar is never traded on its own close.

05 Direction: **long only.** No shorts, no leverage, no hedges.

06 Size: **100% of current equity**, compounded trade to trade. Size never varies by conviction, volatility or stop distance.

07 One position at a time. No pyramiding, no averaging down, no re-entry while a position is open.

EXIT – THE STOP

01 Stop price is fixed at entry: `stop = entry × (1 - 0.078156)` – a **~7.8%** stop (exactly `7.8156%`).

02 The stop is **static**. It never trails, never widens, never moves to breakeven.

03 Checked against each subsequent bar's **low**, starting at bar `t+2` – the entry bar itself is never stop-checked.

04 If `low ≤ stop`, the trade closes **at the stop price** on that bar. Gaps through the level are not modelled.

05 Whichever comes first – stop or trend exit – ends the trade.

EXIT – THE TREND

01 On the close of bar `k`, `SMA20` crosses **below** `SMA50`.

02 Fill at the **close of bar k+1**.

03 A position still open on the last bar of the dataset is closed at that bar's close.

THE CURVE

Equity, trade by trade



Marked at every closed trade. Final equity **23.0x** from 1.00x.

HOW FAR UNDERWATER, AND WHEN

Drawdown



Worst closed-equity drawdown **-30.5%**. This is measured on realised equity between trades, so it understates the intra-trade pain – for that, read the **Worst** column in the trade table.

ALL 36

Every trade



Each bar is one trade, scaled by its return. Above the line is a win; below is a loss, red where the stop ended it.

6	2019-02-22	2019-07-30	158d	$3,937	$3,630	$9,574	TREND	+143.17%	1.69x	-6.8%	+254.8%
7	2019-08-18	2019-08-29	11d	$10,216	$9,418	$9,418	STOP	-7.82%	1.56x	-8.8%	+7.2%
8	2019-11-06	2019-11-12	6d	$9,308	$8,580	$8,580	STOP	-7.82%	1.44x	-8.0%	+1.4%
9	2020-01-09	2020-03-06	57d	$8,055	$7,425	$9,132	TREND	+13.37%	1.63x	-4.8%	+30.4%
10	2020-04-20	2020-06-30	71d	$7,121	$6,565	$9,139	TREND	+28.33%	2.09x	-5.2%	+45.8%
11	2020-07-28	2020-09-03	37d	$11,030	$10,168	$10,168	STOP	-7.82%	1.93x	-9.7%	+13.0%

THE RECOVERY TEST

Did price come back?

For every trade the stop ended, price is checked for the next **3 trading days**: did the daily close get back above the stop level? A "yes" means the stop did not save the trade – it ended a position that was about to recover.

RESULT FOR THIS VERSION

9 of 15 stop-outs (60.0%) came back within 3 days.

THE COMPLETE SWEEP

All 500 stop levels

Every level from 0.5% to 25%, evenly spaced, each one a full re-run over the same 36 trades. Sort any column; the four headline versions are rows in this table, not separate experiments.

BEST +2,198%



Peak at **7.82%** (+2,198%); worst at **1.92%** (-8%).

SHOW All 500 levels FIND filter by stop %...

0.50%	+27%	8.3%	-9.5%	33/36	58%
0.55%	+25%	8.3%	-10.4%	33/36	58%
0.60%	+23%	8.3%	-11.3%	33/36	58%
0.65%	+21%	8.3%	-12.2%	33/36	61%
0.70%	+19%	8.3%	-13.0%	33/36	61%
0.75%	+17%	8.3%	-13.9%	33/36	61%
0.79%	+15%	8.3%	-14.8%	33/36	61%
0.84%	+13%	8.3%	-15.6%	33/36	61%
0.89%	+11%	8.3%	-16.4%	33/36	61%
0.94%	+10%	8.3%	-17.2%	33/36	64%
0.99%	+8%	8.3%	-18.1%	33/36	64%
1.04%	+6%	8.3%	-18.9%	33/36	64%
1.09%	+4%	8.3%	-19.7%	33/36	61%

Showing **500** of 500 levels. Sorted by pct ascending.

WHERE THE RETURN TURNED NEGATIVE

The band that lost money

Of 500 stop levels, **7** finished net negative. They are not scattered – they form a single contiguous band between **1.24%** and **1.92%**. Everywhere else, including a 1% stop, the strategy still made money; it just made far less than it could have.

LEVELS THAT LOST	BAND	WORST LEVEL
7 of 500	**1.24-1.92%**	**1.92%**
1.4% of the sweep	contiguous, not scattered	-8.3% return

Every losing level, in full.

STOP	RETURN	WIN RATE	MAX DD	STOPS FIRED	CAME BACK
1.24%	-0.68%	8.3%	-22.0%	33/36	64%
1.68%	-0.68%	11.1%	-28.7%	32/36	72%
1.73%	-2.25%	11.1%	-29.4%	32/36	75%
1.78%	-3.80%	11.1%	-30.1%	32/36	75%
1.83%	-5.33%	11.1%	-30.8%	32/36	72%
1.87%	-6.83%	11.1%	-31.5%	32/36	72%
1.92%	-8.31%	11.1%	-32.2%	32/36	72%

WHY THIS BAND AND NOT A WIDER ONE

Below ~1.2% the stop is hit so early and so often that the average loss per stop-out is tiny – many trades are cut for less than they cost. Above ~1.9% the surviving winners get large enough to pay for the stop-outs. The losing band is the narrow region where the stop fires on nearly every trade **and** each exit is expensive enough to matter.

THE HONEST READING

"Tight stops lose money" is **too strong** a claim from this data. Only 7 of 500 levels actually went negative. The defensible claim: a tight stop **collapses the return** – +325% against +2,198% – while feeling safer the whole way.

RULE TEST #02 · the full record BTCUSDT, Binance, daily candles Personal experience. Not financial advice.

READ THIS BEFORE COPYING ANYTHING

Method & limits

The result above is real and reproducible from the rules on the strategy tab. It is also narrow. Everything that would weaken it is listed here rather than left for someone else to find.

THE BEST NUMBER ON THIS PAGE IS PARTLY SELECTION

The **7.82% / +2,198%** result is the maximum of 500 tries over the **same 36 trades**. Picking the best of many variants on one small sample is exactly how data-mined edges are manufactured. It is reported here as the peak of a curve, not as a setting to trade – and the curve's **shape** (collapse under 3%, broad plateau from 6–14%) is the finding that survives, not the argmax.

SAMPLE SIZE

- **36 trades.** That is a small sample for any claim about win rate, and the 500 variants are not 500 independent tests – they all reuse the same entries, so their results are heavily correlated.
- Two or three different trades would move every figure here materially.

ONE MARKET, ONE SIGNAL

- BTCUSDT only. Nothing here has been checked on another asset, another timeframe, or another entry rule.
- A 20/50 daily crossover on a high-volatility asset is close to a worst case for a tight percentage stop. On a slower instrument the same test would likely be kinder to tight stops.

COSTS ARE NOT MODELLED

- **No fees, funding or slippage. Stops fill exactly at the stop price.**
- Tight stops trade far more often, so fees and slippage would hurt them **more** than the wide ones. Including costs would widen the gap, not close it.
- Gaps through a stop level are not modelled – real fills below the stop would make tight stops worse again.

A KNOWN ASYMMETRY

- Entries fill at the **open** of the bar after the signal; trend exits fill at the **close** of the bar after the signal.
- That inconsistency is inherited from the run that produced the published figures. Correcting it would change every number on this page, so it is documented instead of silently fixed – and it applies identically to all 500 versions, so the comparison between them stands.

WHAT WOULD ACTUALLY CHANGE THE ANSWER

Re-running on three more assets and two more entry signals, with fees and gap fills modelled, and reporting the **plateau** rather than the peak. If the collapse under 3% survives all of that, the finding is real. Until then it is one honest test of one system on one coin.

REPRODUCING THIS

- Data: `data/btcusdt_daily_2018_2026.json` – raw Binance daily klines.
- Run: `python3 tools/build_rt02_dataset.py` – rebuilds every figure on this page from those bars.
- The rules on the strategy tab are the whole specification; nothing else is applied.